Exhibit 2.3 also illustrates longevity risk by showing the probability of survival to different ages beyond sixty-five. For instance, with SSA data, 91 percent of men are still alive by age seventy (i.e., 9 percent did not live to their seventieth birthday), 63 percent are still alive at eighty, and 22 percent are still alive at ninety. With retirement planning, the trouble is knowing what age to plan for, as this distribution is quite wide.

But more relevant for readers—again because the data reflects people with more similar backgrounds as well as projected improvements in mortality—are the numbers from the SOA data, with mortality improvements projected for a starting year of 2017. The probability of a sixty-five-year-old reaching age ninety-five is 22 percent for males, 29 percent for females, and 45 percent for at least one member of an opposite-gender couple.

In 1994, William Bengen chose thirty years as a conservative planning horizon for a sixty-five-year-old couple when he discussed sustainable retirement spending. But as mortality improves over time, this planning horizon is becoming less conservative. A thirty-year time horizon is not so conservative when we look at data that better reflects higher earning and more highly educated individuals.

Exhibit 2.4 illustrates this longevity risk for a sixty-five-year-old who builds a retirement plan using a thirty-year planning horizon. Again, the probability of outliving this time horizon is not insignificant. The probability of a sixty-five-year-old reaching age ninety-five is 22 percent for male annuitants, 29 percent for female annuitants, and 45 percent for at least one member of an opposite-gender annuitant couple. Longevity risk is the risk of living longer than anticipated and not having the resources to sustain spending for a longer lifetime.

Exhibit 2.4

The Probability of Survival from Age 65 and the Longevity Risk for a Planning Age of 95